FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Respiratory & ENT Examination

Clinical skill | Week 1 · Breathing, fever & the clinical methods | Dr Mawanda, Dr Nabatte

Examining a child's chest, ears, nose and throat is mostly watching before touching - counting breaths for a full minute in a calm child, looking for lower chest wall indrawing, grunting, nasal flaring and head nodding, measuring $SpO_2$, then auscultating opportunistically and leaving the ears and throat until last (and never looking in the throat of a child with stridor and drooling); the findings are interpreted against age-specific norms and turned into a WHO/IMCI classification and a plan.

**Must know before the ward round**

1. Order: observe (from the end of the bed, child on parent's lap) → count RR for 60 s and measure $SpO_2$ → inspect chest → palpate → percuss → auscultate → ENT last (ears, nose, then throat) → lymph nodes → growth.
2. WHO fast breathing: ≥60/min (<2 months), ≥50 (2–11 months), ≥40 (1–4 years); >30 (≥5 years, UCG 2023). Count for a full minute while the child is calm; in infants watch the abdomen.
3. Lower chest wall indrawing: the lower chest wall goes IN when the child breathes IN; it must be present all the time in a calm child. Intercostal soft-tissue recession alone is not indrawing.
4. Signs of severe respiratory distress: grunting, nasal flaring, head nodding, severe indrawing, inability to drink/feed, central cyanosis; $SpO_2$ <90% = hypoxaemia → oxygen.
5. Stridor = inspiratory noise from the larynx/trachea (croup, epiglottitis, foreign body); wheeze = expiratory musical noise from intrathoracic airways (asthma, bronchiolitis). Stridor in a calm child is a danger sign.
6. Never examine the throat with a spatula in a child with stridor, drooling or a toxic look (epiglottitis): you can cause complete obstruction.
7. Otoscopy: largest speculum that fits; pull the pinna DOWN and back in infants, UP and back in older children; brace the otoscope hand on the child's head. Normal drum: pearly grey, translucent, light reflex anteroinferior, malleus visible.
8. AOM = red, bulging drum with lost landmarks (± pus); glue ear = dull, retracted drum with fluid or bubbles; CSOM = perforation with discharge ≥14 days.
9. Tonsil size (Brodsky): 1+ <25%, 2+ 25–50%, 3+ 50–75%, 4+ >75% of the oropharyngeal width; look for exudate, membrane (diphtheria), uvula deviation (quinsy), palatal petechiae.
10. Clubbing in a Ugandan child: think bronchiectasis (post-measles/pertussis/TB), HIV lung disease (LIP), cyanotic heart disease, empyema/lung abscess, cystic fibrosis. Asthma does NOT cause clubbing.
11. Transmitted upper-airway sounds are loudest over the neck, are the same on both sides, and change after the child coughs or the nose is cleared; don't call them crackles.
12. Normal values (APLS): RR <1 y 30–40, 1–2 y 25–35, 2–5 y 25–30, 5–12 y 20–25; HR <1 y 110–160, 1–2 y 100–150, 2–5 y 95–140, 5–12 y 80–120.


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# In a nutshell

The respiratory and ENT examination of a child is the **most frequently performed examination on the Ugandan paediatric ward**: pneumonia, malaria with fast breathing, asthma, bronchiolitis, TB, croup and ear infections fill the beds and the OPD. The examination has two goals:

1. **Recognise the sick child fast**: count the breaths, look for indrawing and danger signs, measure $SpO_2$, and decide whether the child needs oxygen and admission **before** you finish the detailed examination.
2. **Find the cause**: localise the problem (upper airway, lower airway, lung parenchyma, pleura, outside the lung) and examine the ears, nose and throat, which are often the source of fever.

A child is not a small adult: **most of the examination is observation**, done while the child sits on the parent's lap. Auscultation in a crying infant is unreliable; counting, looking and measuring are reliable. That is why WHO/IMCI built pneumonia classification on **respiratory rate, chest indrawing, danger signs and $SpO_2$**, not on crackles.

> **CLINICAL PEARL**
> The one thing never to forget: **look before you touch, count before you listen, and leave the throat until last.** A calm child gives you an accurate RR and a readable $SpO_2$; an upset child gives you neither, and in a child with upper-airway obstruction, upsetting them can be fatal.

# Definitions & classification

| Sign | Definition | What it suggests |
|---|---|---|
| **Fast breathing (tachypnoea)** | RR at or above the WHO age cut-off (≥60 / ≥50 / ≥40; >30 at ≥5 y) | Pneumonia, bronchiolitis, asthma; also acidosis (malaria, DKA, dehydration), fever, heart failure |
| **Lower chest wall indrawing** | Lower chest wall moves **in** on **inspiration** (subcostal) | Stiff lungs or airway obstruction: pneumonia (IMCI "pneumonia"), bronchiolitis, asthma |
| **Recession** | Soft tissues between ribs (intercostal), above the sternum (suprasternal/tracheal tug) or clavicles sink on inspiration | Increased work of breathing; suprasternal recession suggests upper-airway obstruction |
| **Nasal flaring** | Widening of the nostrils with inspiration | Respiratory distress; predicts hypoxaemia in infants |
| **Grunting** | Short noise at the end of expiration (partially closed glottis) | Severe distress: pneumonia, neonatal respiratory distress, sepsis |
| **Head nodding (bobbing)** | Head moves with each breath from sternomastoid use (infants) | Severe distress, hypoxaemia |
| **Central cyanosis** | Blue tongue/inner lips | Hypoxaemia ($SpO_2$ usually <80–85%); may be missed in anaemia |
| **Stridor** | Harsh, mainly **inspiratory** noise from the larynx/upper trachea | Croup, epiglottitis, foreign body, diphtheria, laryngomalacia; **in a calm child = danger sign** |

| Sign | Definition | What it suggests |
|---|---|---|
| **Wheeze** | Musical, mainly **expiratory** noise from intrathoracic airways | Asthma, bronchiolitis, viral wheeze; **monophonic/unilateral** = foreign body or node compression |
| **Crackles (crepitations)** | Discontinuous, non-musical sounds, mainly inspiratory; **fine** (alveoli: pneumonia, oedema, bronchiolitis) or **coarse** (larger airways: secretions, bronchiectasis) | Parenchymal disease or secretions |
| **Bronchial breathing** | Harsh breath sound with a pause between inspiration and expiration, expiration as long as inspiration | **Consolidation**; above an effusion |
| **Pleural rub** | Creaking sound in both phases | Pleurisy |
| **Clubbing** | Loss of the nail-fold angle, increased curvature, fluctuant nail bed; **Schamroth window** lost | Chronic suppurative lung disease, cyanotic heart disease, HIV/LIP, CF, IBD |
| **Hyperinflation** | Barrel chest, increased AP diameter, liver pushed down, reduced cardiac dullness | Asthma, bronchiolitis |
| **Harrison sulcus** | Horizontal groove along the lower ribs at the diaphragm insertion | Long-standing increased work of breathing (chronic asthma, rickets, heart failure) |

### WHO/IMCI classification the examination feeds into (2–59 months)

| Findings | Classification |
|---|---|
| Danger sign, $SpO_2$ <90%/central cyanosis, severe distress, stridor in a calm child | **Severe pneumonia or very severe disease** |
| Fast breathing and/or chest indrawing | **Pneumonia** |
| None of these | **Cough or cold** |

### Normal vital signs by age (APLS, as used on Ugandan wards)

| Age | Respiratory rate (/min) | Heart rate (/min) | Systolic BP (mmHg) |
|---|---|---|---|
| Newborn | 40–60 | 110–170 | 60–80 |
| <1 year | 30–40 | 110–160 | 70–90 |
| 1–2 years | 25–35 | 100–150 | 80–95 |
| 2–5 years | 25–30 | 95–140 | 80–100 |
| 5–12 years | 20–25 | 80–120 | 90–110 |
| >12 years | 15–20 | 60–100 | 100–120 |

Minimum acceptable SBP ≈ **70 + (2 × age in years)** for children 1–10 years. **$SpO_2$ normal ≥95%** at Kampala's altitude (about 1,200 m); **<90% = hypoxaemia** needing oxygen.

# Equipment

| Item | Use / tip |
|---|---|
| **Watch or respiratory timer** (IMCI timer, phone) | Count RR for 60 s |
| **Pulse oximeter** with neonatal/paediatric/adult probes | Correct probe size; finger, toe, palm or foot in infants |

| Item | Use / tip |
|---|---|
| **Stethoscope** with a **small (paediatric) diaphragm and bell** | Warm it first; small chest = small diaphragm |
| **Otoscope** with charged batteries and **specula of several sizes** (disposable or cleaned between children) | Largest speculum that fits the canal |
| **Pen torch** and **wooden tongue depressors** | Throat; one depressor per child |
| **Tape measure, weighing scale, length board, MUAC tape** | Growth (Z-scores) |
| **Thermometer** (axillary digital) | Fever |
| **Peak flow meter** with mouthpieces and a height chart | Children ≥5–6 years with asthma |
| **Tuning fork 512 Hz** | Rinne and Weber in older children |
| **Gloves, alcohol hand rub, clean specula** | Infection prevention between children |
| **Toys, stickers or a phone video** | Distraction |

# Indications & contraindications

### Indications

- Every child with cough, difficult or fast breathing, noisy breathing, fever, ear pain or discharge, sore throat, neck swelling, poor feeding, or failure to thrive.
- Every admitted child daily (RR, $SpO_2$, work of breathing), and every child with HIV, TB contact, malnutrition or sickle cell disease at each visit.

### Contraindications / cautions

- **Do not examine the throat** with a spatula (and avoid lying the child down, IM injections or painful procedures) in a child with **stridor, drooling, tripod posture or toxic appearance**: suspected **epiglottitis**, retropharyngeal abscess or severe croup. Secure the airway first.
- **Do not delay resuscitation** to complete the examination: a child with emergency signs gets oxygen and ABC care first.
- **Otoscopy**: don't insert the speculum blindly or force it in a struggling child; avoid if you suspect a canal **foreign body** that you might push further (refer for removal with proper instruments).
- **Percussion and tactile fremitus** are of limited value and can upset infants: do them in older children or when you suspect an effusion.

# Procedure step by step

## A. Preparation and approach

1. **Wash your hands** (alcohol rub) and introduce yourself to the carer and child. Explain what you will do; ask permission.
2. Note the **age** (it sets every normal value), **weight** and whether the child is **asleep, calm, crying or feeding**.
3. Position: **infants and toddlers on the parent's lap**, facing you or sideways; **older children** sitting on the bed, then lying at 45°.
4. **Undress the chest** fully (vest off), keeping the child warm; in older girls and adolescents, offer privacy and a chaperone.
5. **Warm** your hands and stethoscope; use a toy or the parent's voice to distract.

6. **Opportunistic order**: do the quiet things first (observe, count, auscultate if asleep), and the upsetting things last (ears, throat, temperature in some children).

### B. General inspection (from the end of the bed)

- **Well or unwell?** Alert, playing, feeding, smiling vs lethargic, irritable, floppy.
- **Posture**: tripod/sniffing (epiglottitis), sitting forward (asthma), lying still (pleuritic pain).
- **Colour**: central cyanosis (tongue), pallor (palms, conjunctivae), jaundice.
- **Audible noises**: stridor, wheeze, grunting, cough (barking = croup; paroxysmal with whoop = pertussis; wet = secretions/bronchiectasis; staccato in young infants = chlamydia).
- **Work of breathing**: nasal flaring, head nodding, tracheal tug, indrawing, abdominal breathing, use of accessory muscles.
- **Ability to talk/feed** (asthma severity, respiratory distress).
- **Drooling, mouth breathing**, voice (hoarse in croup, muffled in epiglottitis/quinsy).
- **Nutritional status**: wasting, oedema, hair changes; measure weight, length/height, MUAC; plot Z-scores.
- **Equipment around the bed**: oxygen, nebuliser, inhalers, sputum pot, NG tube, drains.

### C. Vital signs

1. **Respiratory rate**: child calm, chest exposed; watch the **abdomen** in infants (diaphragmatic breathing); count for **a full 60 seconds**; repeat if borderline or if the child was crying. State the result against the WHO cut-off ("RR 54: fast for a 9-month-old").
2. **Pulse oximetry**: choose the right probe; place on a finger/toe (big toe or foot in infants, wrapped in the neonatal sensor); keep the limb still and warm; wait for a **steady pleth waveform** and a stable reading for **30–60 seconds**; compare the pulse rate with your own count. Note **in air** or on oxygen (and the flow).
3. **Heart rate**, **capillary refill** (press the sternum or fingertip for 5 s; normal ≤2 s; >3 s abnormal), **temperature** (axillary), **BP** (appropriate cuff: bladder covers 80–100% of arm circumference) when unwell.

> CLINICAL PEARL
> **Pulse oximeters can over-read in children with dark skin** (by 2–3% on average) and when perfusion is poor or the child moves. Treat a borderline reading (90–93%) in a child with signs of distress as possible hypoxaemia, repeat it, and watch the child, not only the number.

### D. Hands, face and neck

- **Hands**: **clubbing** (look along the nail from the side; Schamroth window test in older children), peripheral cyanosis, warmth, CRT, palmar pallor, flapping tremor ($CO_2$ retention, rare in children), salbutamol tremor.
- **Face**: central cyanosis (tongue), nasal flaring, conjunctival pallor, **allergic shiners and nasal crease** (atopy), facial puffiness, parotid swelling (HIV), periorbital swelling (sinusitis, nephrotic).
- **Neck**: tracheal position (older children; gently, the trachea should be central in the suprasternal notch), tracheal tug, **lymph nodes** (see ENT), JVP (older children).

### E. Chest: inspection

1. **Shape**: normal, barrel (hyperinflation), **pectus excavatum/carinatum**, **Harrison sulci**, rickety rosary (rickets), asymmetry (collapse, effusion, scoliosis).
2. **Scars**: chest drain sites, thoracotomy, BCG scar (right upper arm in Uganda).
3. **Movement**: symmetry (reduced on the side of pathology), **lower chest wall indrawing** (state whether it is present all the time in a calm child), intercostal, subcostal, suprasternal recession, paradoxical (see-saw) breathing.

4. **Respiratory pattern**: rate, depth (deep acidotic breathing in malaria/DKA/dehydration), apnoeas (count pauses >20 s in infants), periodic breathing (preterms).

### F. Chest: palpation

1. **Trachea** (older children) and **apex beat** (mediastinal shift: deviated away from effusion/pneumothorax, towards collapse).
2. **Chest expansion** (children ≥5 y): hands around the lower chest, thumbs meeting in the midline, ask for a deep breath; normal ≥3–5 cm; unequal = pathology on the side that moves less.
3. **Tactile vocal fremitus** (older children: "say ninety-nine"; infants: palpate during crying): increased over consolidation, decreased over effusion/pneumothorax.
4. Feel for **subcutaneous emphysema** (crackling under the skin of the neck/chest: pneumothorax, pneumomediastinum in asthma) and **tenderness**.

### G. Chest: percussion

- Use light percussion (one finger directly on the chest in infants; standard technique in older children), comparing **side to side**, front, sides (axillae) and back.
- **Resonant** = normal; **hyper-resonant** = pneumothorax, hyperinflation; **dull** = consolidation, collapse; **stony dull** = **pleural effusion/empyema**.
- Liver dullness normally begins at the 5th–6th rib in the mid-clavicular line; lower = hyperinflation.

### H. Chest: auscultation

1. Use the **small diaphragm**, warmed; listen when the child is quiet (asleep on the parent, or after distraction); in older children ask them to breathe in and out through the mouth.
2. Listen **symmetrically**: front (upper, middle, lower zones), axillae (essential: the right middle lobe and lingula are heard there), back (upper, lower), comparing left and right at each level.
3. Describe:
    - **Breath sounds**: vesicular (normal) vs **bronchial** (consolidation); **reduced/absent** (effusion, pneumothorax, collapse, silent chest).
    - **Added sounds**: fine/coarse **crackles** (inspiratory, early or late), **wheeze** (expiratory, polyphonic vs monophonic), **stridor** (listen over the neck), **pleural rub**.
    - **Vocal resonance** in older children (increased over consolidation, absent over effusion).
4. **Transmitted upper-airway sounds** are loudest over the trachea/neck, symmetrical, and change after a cough or after clearing the nose: listen over the neck and ask the child to cough (or suction the nose) before calling them crackles or wheeze.
5. Finish by listening to the **heart** (murmurs, gallop in heart failure, muffled sounds in pericardial effusion) and palpating the **liver** (hepatomegaly in heart failure; pushed down in hyperinflation).

### I. Ears

1. **Inspect** the pinna and **behind the ear** (mastoid: redness, swelling, tenderness, pinna pushed forward and down = mastoiditis), pre-/post-auricular nodes, discharge, smell.
2. **Position**: child on the parent's lap, sideways, one of the parent's arms around the child's body and arms, the other hand holding the head against the parent's chest.
3. **Otoscope**: choose the **largest speculum** that fits; hold the otoscope like a pen, **brace the little finger on the child's head** so the otoscope moves with the head.
4. **Straighten the canal**: **infants – pull the pinna gently down and back; older children – up and back.**
5. Look at the **canal** (wax, foreign body, swelling and pain on moving the pinna = otitis externa, discharge) and the **tympanic membrane**:

| Finding | Normal | AOM | OME (glue ear) | CSOM |
|---|---|---|---|---|
| Colour | Pearly grey, translucent | **Red/yellow, opaque** | Dull, amber/grey | Varies; discharge |
| Position | Neutral | **Bulging** | **Retracted** (prominent malleus handle) | – |
| Light reflex | Anteroinferior cone | Lost | Distorted | – |
| Landmarks (malleus) | Visible | **Lost** | Prominent | – |
| Other | Mobile (pneumatic) | Reduced mobility; pus if perforated (<14 days) | **Fluid level, bubbles**, reduced mobility | **Perforation** (central or attic) with discharge **≥14 days**; cholesteatoma = white debris, foul smell |

6. **Hearing**: infants – **distraction test** (sound behind and to the side; child turns), parental concern, speech milestones; older children – **whisper test** (masking the other ear), **Rinne** (512 Hz fork: air conduction louder than bone = normal or sensorineural loss; bone > air = conductive loss) and **Weber** (lateralises to the conductive-loss ear).

## J. Nose

1. **Inspect** externally: shape, nasal crease (allergic salute), flaring, discharge (clear, purulent, **unilateral foul** = foreign body; bloody), vestibulitis.
2. **Anterior rhinoscopy**: push the tip up gently with your thumb and look with the otoscope (large speculum) or torch: septum (deviation, haematoma after trauma, perforation), **inferior turbinates** (pale, swollen = allergic rhinitis; red = infection), **polyps** (grey, glistening; think **cystic fibrosis** in a child), foreign body.
3. **Patency**: hold a cold metal spatula or mirror under each nostril (misting), or watch a wisp of cotton wool move; important in neonates (choanal atresia: pass a fine catheter).
4. **Sinuses** (older children): tenderness over maxillary and frontal sinuses (frontal sinuses develop after about 7 years); periorbital swelling.

## K. Mouth and throat (leave until last)

> **RED FLAG**
> **Stop.** If the child has **stridor, drooling, a muffled voice, tripod posture or looks toxic**, do NOT examine the throat. Say so out loud in an OSCE, call for senior/anaesthetic help, and keep the child calm with the parent.

1. **Position**: child on the parent's lap facing forward; parent's one arm across the child's arms and chest, other hand on the forehead holding the head against the parent's chest.
2. Ask older children to **open wide and say "aah"** (or "show me your lion roar"); many children will open without a spatula.
3. Use a **torch** and, if needed, a **tongue depressor** placed on the **anterior two-thirds of the tongue** (not the back: it triggers gagging); press down briefly.
4. Look at:
   - **Lips and mouth**: cyanosis, angular stomatitis, cheilitis, **oral thrush** (HIV, antibiotics), ulcers (herpes, HFMD), **Koplik spots** (measles: white spots on the buccal mucosa opposite the molars), gums (gingivitis, bleeding), teeth (caries, dental abscess), tongue (strawberry tongue in scarlet fever/Kawasaki).
   - **Palate**: petechiae (GAS, EBV), cleft, bulging soft palate (quinsy).

- **Tonsils**: size (Brodsky grade), symmetry, redness, **exudate** (white/yellow spots), **membrane** (grey, adherent, bleeds when removed = **diphtheria**), ulcers.
- **Uvula**: central or **deviated away** from a quinsy.
- **Posterior pharyngeal wall**: redness, lymphoid follicles, bulging (retropharyngeal abscess), post-nasal drip.

**Brodsky tonsil grading**

| Grade | Tonsils occupy |
|---|---|
| 0 | Within the tonsillar fossa (or removed) |
| 1+ | <25% of the oropharyngeal width |
| 2+ | 25–50% |
| 3+ | 50–75% |
| 4+ | >75% ("kissing tonsils") |

## L. Lymph nodes of the head and neck

1. Stand **behind** an older child (in front for a toddler on the parent's lap), head slightly flexed.
2. Palpate in order: **submental → submandibular → tonsillar (jugulodigastric) → anterior cervical chain → posterior cervical chain → supraclavicular → pre-auricular → post-auricular → occipital**.
3. For each node: **site, size** (cervical nodes up to 1 cm are common and normal in children; >2 cm, or any **supraclavicular** node, needs explanation), **consistency** (soft, rubbery, firm, hard), **tenderness**, **mobility**, **matting**, **fluctuation**, overlying skin (redness, **sinus**).
4. Interpretation: **tender anterior cervical nodes** with tonsillitis = GAS; **posterior cervical** with splenomegaly = EBV; **firm, matted, non-tender, ± fluctuant or sinus** = **TB lymphadenitis**; **generalised lymphadenopathy** = HIV, TB, EBV, leukaemia/lymphoma; **hard, fixed, rapidly growing** = malignancy (Burkitt, Hodgkin).
5. Check the **axillae** (BCG adenitis on the vaccinated side) and **groin**, and feel for **hepatosplenomegaly** if nodes are generalised.

## M. Completing the examination

- **Peak flow** (≥5–6 y with asthma): standing, best of three.
- **Growth**: weight, height/length, MUAC; Z-scores.
- **Other systems** guided by findings: heart (murmurs, gallop), abdomen (liver, spleen), CNS (meningism in a febrile child), skin (rash, eczema).
- **Thank** the child and carer, cover the child, wash hands, clean the otoscope specula.
- **Summarise** findings to the examiner/senior (see section N).

## N. Presenting your findings

> "On general examination, Amina is a 2-year-old girl, alert, sitting on her mother's lap, not cyanosed or pale, weighing 11 kg (WAZ −0.8). Her respiratory rate is 48 per minute counted over one minute, which is fast for her age; $SpO_2$ is 93% in room air; heart rate 136; temperature 38.6 °C. She has nasal flaring and lower chest wall indrawing but no grunting or head nodding. The trachea is central. Chest expansion appears reduced on the right. Percussion is dull at the right lower zone posteriorly, with bronchial breathing and fine crackles there; the left side is clear; no wheeze. Ears: both drums pearly grey with a normal light reflex. Throat: mildly red, tonsils grade 1 without exudate. Small, soft, mobile cervical nodes under 1 cm. No clubbing, no hepatosplenomegaly. **In summary**, she has fast breathing, chest indrawing and signs of right lower lobe consolidation without danger signs or hypoxaemia: **pneumonia (WHO classification)**, right lower lobe."

# OSCE checklist

| # | Step | Done ? |
|---|---|---|
| 1 | Washes hands; introduces self; confirms child's name and age; explains and gets consent from carer (and child) | ☐ |
| 2 | Positions child appropriately (parent's lap for infants/toddlers); exposes chest; keeps warm; warms hands and stethoscope | ☐ |
| 3 | General inspection from the end of the bed: well/unwell, posture, colour, noises, drooling, feeding/talking | ☐ |
| 4 | **Counts RR for 60 seconds** with the child calm and **states the WHO age cut-off** | ☐ |
| 5 | **Measures $SpO_2$** with the correct probe and waits for a stable waveform; states in air/oxygen | ☐ |
| 6 | Checks HR, CRT and temperature; comments on growth (weight, Z-score/MUAC) | ☐ |
| 7 | Looks for **danger signs**: unable to drink/feed, lethargy, central cyanosis, grunting, head nodding, nasal flaring | ☐ |
| 8 | Inspects hands (clubbing, cyanosis), face (flaring, pallor, shiners), neck (tracheal tug, nodes) | ☐ |
| 9 | Inspects the chest: shape, symmetry, scars, **lower chest wall indrawing** (defined correctly) | ☐ |
| 10 | Palpates trachea/apex (older child), expansion, tactile fremitus where appropriate | ☐ |
| 11 | Percusses symmetrically front, axillae and back | ☐ |
| 12 | Auscultates symmetrically front, **axillae** and back; identifies breath sounds and added sounds; checks for transmitted sounds | ☐ |
| 13 | States that the **throat will not be examined if there is stridor/drooling** | ☐ |
| 14 | Otoscopy: correct speculum, **pinna down/back (infant) or up/back (older)**, braces hand, describes canal and drum | ☐ |
| 15 | Examines nose (discharge, turbinates, polyps, foreign body, patency) | ☐ |
| 16 | Examines mouth and throat safely: tongue depressor on anterior tongue, tonsils graded, exudate/membrane/uvula noted, Koplik spots/thrush | ☐ |
| 17 | Palpates head and neck nodes in order and describes size, consistency, tenderness, matting | ☐ |
| 18 | Mentions completing with heart, liver/spleen, growth chart, PEF (≥5 y) | ☐ |
| 19 | Thanks child and carer, covers child, washes hands, cleans equipment | ☐ |
| 20 | Presents findings concisely with a **WHO classification** and a differential | ☐ |

# Clinical acumen

### Focused history (why each question matters before you examine)

- **Age** → normal RR/HR, fast-breathing cut-off, likely diagnoses (bronchiolitis <2 y; croup 6 m–3 y; GAS ≥3 y).
- **Noisy breathing: when, what sound?** Inspiratory (stridor) vs expiratory (wheeze); sudden (foreign body) vs gradual.
- **Danger signs** (drink, vomit, convulsions, lethargy) → severe disease regardless of your chest findings.
- **Fever pattern and duration** → bacterial vs viral; >2 weeks → TB.
- **Ear discharge duration** (<14 vs ≥14 days) → acute vs chronic.
- **Drooling, voice change** → epiglottitis, quinsy (don't examine the throat).
- **HIV status, TB contact, immunisations, nutrition** → change the meaning of your findings (clubbing, nodes, thrush).

## Examination: the key findings and their meaning

| Finding | Meaning |
|---|---|
| RR above cut-off, no indrawing | Pneumonia (WHO), or fever/acidosis/wheeze: classify after excluding wheeze and malaria acidosis |
| Lower chest wall indrawing | Pneumonia (WHO); severe if very marked |
| $SpO_2$ <90%, grunting, head nodding | Severe pneumonia/hypoxaemia: oxygen now |
| Stridor at rest | Upper-airway obstruction: croup, epiglottitis, foreign body, diphtheria |
| Diffuse polyphonic wheeze | Asthma, viral wheeze, bronchiolitis |
| Unilateral/monophonic wheeze | Foreign body, TB node compression |
| Fine crackles bilaterally in an infant | Bronchiolitis, pneumonia, heart failure |
| Dullness + bronchial breathing | Consolidation |
| Stony dullness + absent breath sounds + trachea shifted away | Effusion/empyema |
| Hyper-resonance + absent sounds + trachea shifted away | Pneumothorax (tension if shocked) |
| Dullness + reduced sounds + trachea pulled towards | Collapse |
| Clubbing | Chronic lung suppuration, HIV/LIP, cyanotic heart disease, CF |
| Red bulging drum | AOM |
| Perforation + discharge ≥14 days | CSOM |
| Exudative tonsils + tender anterior nodes, no cough | GAS tonsillitis |
| Grey adherent membrane, bull neck | Diphtheria |
| Uvula deviated, trismus | Quinsy |
| Firm matted nodes, sinus | TB lymphadenitis |

## Red flags

> **RED FLAG**
> Stop the routine examination and act if you find: **stridor at rest, drooling or tripod posture**; **$SpO_2$ <90% or central cyanosis**; **grunting, head nodding, severe indrawing, apnoea**; **silent chest** in a wheezy child; **tracheal deviation with shock** (tension pneumothorax); **lethargy or inability to drink**; **mastoid swelling, neck stiffness or facial palsy** with an ear infection; **grey membrane** in the throat.

## Thinking like a clinician

- **Count, don't estimate**: the most common OSCE and ward error is a guessed RR.
- **Indrawing must be seen in a calm child**: a crying child "indraws" normally.
- **Wheeze + fast breathing**: give a bronchodilator and re-count before calling it pneumonia (WHO/IMCI).
- **Deep, fast breathing with a clear chest** = **acidosis** (severe malaria, DKA, dehydration, salicylates), not pneumonia.
- **Crackles + hepatomegaly + murmur** = heart failure (VSD, RHD); don't just give antibiotics.
- **Silence can be dangerous**: a "quiet chest" in asthma or "softening stridor" in croup may mean exhaustion.
- **Unilateral signs in a toddler without fever** = foreign body until proven otherwise.

- **Fever without a source**: always look in the ears and throat, and do a malaria test.
- **Clubbing is never normal**: it changes the differential to chronic disease (bronchiectasis, HIV, TB sequelae, heart disease).
- **Nodes**: a single firm matted cervical node in a child with a TB contact is TB until an FNA says otherwise.

# Clinical workup

For this skill topic the workup is the **pre-examination assessment** (is it safe to examine fully, and in what order?) and the **post-examination checks** (what do the findings mean, what do I do next?).

### Step 0: First 5 minutes

1. **ETAT triage at the door**: Airway/Breathing (obstructed breathing, central cyanosis, severe respiratory distress), Circulation (cold hands + CRT >3 s + weak fast pulse), Coma/Convulsions, Dehydration (severe). If any emergency sign → **resuscitate first** (oxygen, airway positioning, IV access, glucose), examine later.
2. **Priority signs** (3TPR-MOB): tiny baby (<2 months), temperature (very high), trauma, pallor, poisoning, pain (severe), respiratory distress, restless/irritable/lethargic, referral, malnutrition (visible severe wasting), oedema of both feet, burns → examine immediately.
3. **Airway caution**: stridor/drooling → no throat exam, no lying down, call help.
4. **$SpO_2$ within the first minutes** in any child with respiratory symptoms.
5. **Infection control**: if TB or pertussis is suspected in an adolescent/adult carer, masks and ventilation; hand hygiene before and after.

### Step 1: Focused history

- [ ] Age and weight → normal values, drug doses.
- [ ] Main symptom and duration → acute vs chronic pathway.
- [ ] Danger signs → severe classification.
- [ ] Noisy breathing description → upper vs lower airway.
- [ ] Choking episode → foreign body.
- [ ] Ear discharge duration, hearing, speech → AOM/CSOM, hearing loss.
- [ ] HIV, TB contact, immunisations (PCV, Hib, DTP, MR), nutrition → context.

### Step 2: Focused examination

As in **Procedure step by step**: observe → RR and $SpO_2$ → hands/face/neck → chest (inspect, palpate, percuss, auscultate) → ears → nose → throat (if safe) → nodes → growth, heart, abdomen.

| Document these positives | And these negatives |
|---|---|
| RR (60 s) vs cut-off; $SpO_2$ (air/$O_2$) | "No danger signs; no central cyanosis" |
| Indrawing, flaring, grunting, head nodding | "No stridor in a calm child" |
| Focal chest signs (side, zone) | "Air entry equal; no wheeze" |
| Drum appearance each side | "No mastoid swelling" |
| Tonsil grade, exudate, nodes | "No membrane; uvula central; no drooling" |
| Clubbing, growth Z-scores | "No hepatosplenomegaly" |

## Step 3: Bedside tests

| Test | How | What it tells you |
|---|---|---|
| **Pulse oximetry** | Correct probe; 30–60 s stable trace | <90% = oxygen |
| **Respiratory timer** | 60 s count | Fast breathing |
| **Peak expiratory flow** | ≥5–6 y; standing; best of 3; compare with height chart | Asthma severity and response |
| **Bronchodilator trial** | Salbutamol via spacer, re-assess after 15–20 min | Reversible wheeze vs pneumonia |
| **Malaria RDT, RBS, HemoCue** | Finger prick | Common co-existing problems |
| **Tuning fork tests** | 512 Hz Rinne/Weber | Conductive vs sensorineural hearing loss |
| **Throat swab / RADT** (if available) | Swab both tonsils and posterior pharynx | GAS |

## Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Chest X-ray** | Severe pneumonia, unilateral signs, suspected effusion, pneumothorax, foreign body, TB, not improving | Consolidation, effusion, pneumothorax, hyperinflation, hilar nodes, foreign body | Confirms what the examination suggested; drains, bronchoscopy, TB treatment. HC IV (some), hospitals |
| **Chest ultrasound** | Dullness: effusion vs consolidation | Fluid, septations | Tap/drain site. RRH/NRH |
| **Lateral neck X-ray** (stable child only) | Retropharyngeal abscess, radiopaque foreign body | Prevertebral widening, foreign body | ENT/surgery |
| **FBC** | Severe infection, suspected malignancy (nodes) | Neutrophilia, blasts, atypical lymphocytes | Antibiotics, oncology referral |
| **Ear swab** | CSOM not drying | Pseudomonas, Staph | Targeted drops |
| **FNA of node** | Persistent node >2 cm, matted, sinus | Caseation, AFB, Xpert; lymphoma cells | TB treatment vs oncology |
| **Xpert MTB/RIF Ultra** | Chronic cough, TB contact, nodes | MTB detected | TB treatment |
| **Audiometry/tympanometry** | Hearing concern, glue ear, CSOM | Conductive loss; flat tympanogram | ENT (Mulago, Mengo ENT clinics) |
| **HIV test** | Clubbing, thrush, generalised nodes, CSOM, recurrent pneumonia | Positive | ART, cotrimoxazole |

## Step 5: Putting it together

### From signs to localisation

| Pattern | Localisation | Likely diagnosis |
|---|---|---|
| Stridor, barking cough, hoarse voice | Larynx/subglottis | Croup |
| Stridor, drooling, toxic, no cough | Supraglottis | Epiglottitis |

| Pattern | Localisation | Likely diagnosis |
|---|---|---|
| Polyphonic expiratory wheeze, hyperinflation | Small airways (bilateral) | Asthma / viral wheeze |
| Fine crackles + wheeze in an infant | Bronchioles | Bronchiolitis |
| Focal dullness, bronchial breathing, crackles | Parenchyma | Lobar pneumonia |
| Stony dullness, absent sounds, shift away | Pleura | Effusion / empyema |
| Deep fast breathing, clear chest | Outside the lung | Acidosis (malaria, DKA, dehydration) |

EXAM FAVOURITE

**Worked example 1: interpreting a set of chest signs.** A 6-year-old: RR 34, $SpO_2$ 92%, trachea deviated to the left, reduced expansion on the right, **stony dull** percussion right base, **absent breath sounds** and absent vocal resonance there, bronchial breathing just above the dull area. **Interpretation**: a **right pleural effusion** (mediastinum pushed away; bronchial breathing above the fluid from compressed lung). With 3 weeks of fever and a TB contact → TB effusion vs empyema → CXR/ultrasound and diagnostic tap.

EXAM FAVOURITE

**Worked example 2: reading two eardrums.** A 15-month-old with fever and irritability. Right drum: **red, bulging, no light reflex, malleus not seen → acute otitis media**. Left drum: pearly grey, light reflex present → normal. Plan: amoxicillin 40 mg/kg twice daily for 5 days, analgesia, review in 5 days. A 5-year-old with 2 months of painless discharge: canal full of pus; after dry mopping, a **central perforation** is seen → **CSOM**: wicking + ciprofloxacin ear drops for 2 weeks, HIV test, hearing assessment.

# Management

The examination itself is not a treatment, but **what you do with each finding** is the reason you examine. This section covers acting on abnormal findings, the emergency flow, and aftercare.

## 1. Acting on abnormal findings at the bedside

| Finding | Immediate action |
|---|---|
| **$SpO_2$ <90%** or central cyanosis | **Oxygen** by nasal prongs (0.5–1 L/min neonates; 1–2 L/min infants; 2–4 L/min older children), target ≥90% (≥94% if emergency signs); inform senior; classify as severe |
| **Fast breathing ± indrawing, no danger signs** | Classify **pneumonia** → oral amoxicillin 40 mg/kg twice daily × 5 days (see LRTI topic) |
| **Danger sign / severe distress** | **Severe pneumonia** → admit, oxygen, ampicillin 50 mg/kg 6-hourly + gentamicin 7.5 mg/kg daily |
| **Wheeze with fast breathing** | Salbutamol via spacer (2–4 puffs) and re-count in 15–20 min |
| **Stridor at rest** | Keep calm, oxygen without distress, dexamethasone 0.15–0.6 mg/kg (croup), nebulised adrenaline if severe; epiglottitis → call anaesthetist/ENT, no throat exam |
| **Silent chest / exhaustion in asthma** | Life-threatening asthma: back-to-back salbutamol + ipratropium, IV magnesium, senior help |
| **Tension pneumothorax** (shock + tracheal deviation + hyper-resonance) | **Needle decompression** (2nd intercostal space, mid-clavicular line, or 4th–5th ICS anterior axillary line), then chest drain |
| **Stony dullness** | CXR/ultrasound, diagnostic tap, chest drain if pus |
| **AOM** | Amoxicillin 40 mg/kg twice daily × 5 days (WHO/IMCI), paracetamol 15 mg/kg |

| Finding | Immediate action |
|---|---|
| **CSOM** | Teach wicking, ciprofloxacin drops 2 weeks, HIV test, hearing check |
| **Mastoid swelling** | Admit; ceftriaxone 50–80 mg/kg + metronidazole; urgent ENT |
| **Exudative tonsillitis (McIsaac 4–5)** | Penicillin V 10–20 mg/kg 6-hourly × 10 days or benzathine penicillin once |
| **Grey membrane** | Isolate, diphtheria antitoxin, penicillin/erythromycin, airway watch |
| **Foreign body in ear/nose** | Don't push it in; remove only if easily grasped with proper instruments (crocodile forceps, hook) and a still child; otherwise ENT. **Button batteries** in the nose/ear = urgent removal |
| **Wax impaction** | Softening drops (olive oil/sodium bicarbonate) for a few days; no cotton buds |
| **Matted cervical nodes** | TB workup (Xpert on FNA, CXR, HIV) |
| **Clubbing** | Look for chronic lung disease, HIV, heart disease, CF; CXR, HIV test, echo |

### 2. Emergency flow: you find a dangerous sign during bedside teaching

1. **Stop teaching**; tell the tutor/nurse clearly: "This child has $SpO_2$ 84% / stridor at rest / a silent chest."
2. **Call for help** and move the child to the resuscitation area if needed.
3. **ABC**: position, oxygen, suction if needed (not in epiglottitis), IV access, glucose.
4. **Specific first treatment** (as in the table above).
5. **Document** time, findings and actions; hand over to the team.

### 3. Aftercare following examination

- Re-dress and comfort the child; thank the carer and explain findings in plain language.
- **Clean equipment**: otoscope specula (disposable or alcohol-cleaned), stethoscope diaphragm (alcohol wipe) between patients; **hand hygiene** (WHO 5 moments).
- **Document** all findings with times, RR counts and $SpO_2$ (in air/$O_2$), and the classification.
- **Plan re-examination**: severe pneumonia RR and $SpO_2$ at least 3-hourly; asthma every 20–30 min in the first hour; croup after steroid/adrenaline.

### 4. Model case presentations

**Asthma**: "RR 38, HR 138, $SpO_2$ 89% in air, speaks in words, intercostal recession, widespread polyphonic expiratory wheeze, equal air entry, PEF 40%: acute severe asthma."

**Bronchiolitis**: "4-month-old, RR 62, $SpO_2$ 89%, nasal flaring, subcostal recession, bilateral fine inspiratory crackles and expiratory wheeze, hyperinflated chest, liver edge 3 cm below the costal margin (pushed down): severe bronchiolitis."

**Croup**: "Barking cough, hoarse cry, inspiratory stridor at rest, moderate suprasternal and subcostal recession, alert, $SpO_2$ 95%, no drooling: moderate croup; throat not examined."

## Complications

Complications of the **examination** itself (and how to avoid them):

| Complication | Prevention |
|---|---|
| **Complete airway obstruction** from throat examination or lying down in epiglottitis/severe croup | Never examine the throat with stridor/drooling; keep upright; call airway help |

| Complication | Prevention |
|---|---|
| **Inaccurate findings** (RR, $SpO_2$, indrawing) from an upset child | Examine calm child first, opportunistically, on the parent's lap |
| **Ear canal trauma, bleeding, TM perforation** | Largest speculum that fits, brace hand, gentle traction, still child |
| **Pushing a foreign body deeper** | Look first; refer if not easily removable |
| **Gagging, vomiting, aspiration** | Tongue depressor on the anterior tongue, briefly |
| **Cross-infection** (otoscope specula, stethoscope, hands) | Hand hygiene; clean equipment between children |
| **Hypothermia** in neonates/small infants during undressing | Examine under a warmer/blanket, expose briefly |
| **Distress and loss of trust** | Explain, involve parent, distraction, praise |

# Ward-round & exam pearls

**Model case presentation (child):**

> "Amina is a 2-year-old girl from Kawempe weighing 11 kg (WAZ −0.8, WHZ −0.5), fully immunised per UNEPI including PCV ×3 and MR1, HIV-negative, breastfed to 18 months, now on a family diet, with 3 days of fever and cough. On examination she is alert on her mother's lap, not cyanosed, RR 48 counted over a minute (fast for her age; cut-off 40), $SpO_2$ 93% in air, HR 136, T 38.6 °C, with nasal flaring and lower chest wall indrawing, no grunting or head nodding. Trachea central. Dull percussion, bronchial breathing and fine crackles at the right base posteriorly; left clear; no wheeze. Ears and throat normal; small soft cervical nodes. No clubbing, no hepatosplenomegaly. Malaria RDT negative. Assessment: **pneumonia (WHO), right lower lobe, without danger signs or hypoxaemia**. Plan: oral amoxicillin 440 mg (40 mg/kg) twice daily for 5 days, paracetamol, home care advice, review in 2–3 days or earlier if danger signs."

## Likely viva questions with answers

1. *How do you count the respiratory rate in an infant?* Calm child, chest/abdomen exposed, watch the abdomen, count a full 60 s with a timer, repeat if borderline.
2. *Define lower chest wall indrawing.* The lower chest wall goes in when the child breathes in, present all the time in a calm child.
3. *How do you differentiate stridor from wheeze?* Stridor is inspiratory, from the extrathoracic airway, loudest over the neck; wheeze is expiratory, musical, from intrathoracic airways, heard over the chest.
4. *How do you straighten the ear canal in an infant vs an older child?* Pull the pinna down and back in infants; up and back in older children.
5. *What does a normal eardrum look like?* Pearly grey, translucent, with a cone of light anteroinferiorly and the handle of the malleus visible.
6. *When must you not examine the throat?* Stridor, drooling, tripod posture, muffled voice or toxic look: suspected epiglottitis.
7. *Causes of clubbing in a Ugandan child?* Bronchiectasis (post-measles/pertussis/TB), HIV lung disease (LIP), cyanotic congenital heart disease, empyema/lung abscess, CF, IBD, liver cirrhosis, infective endocarditis.

## OSCE tips

- **Say what you are doing and why** ("I am counting for a full minute because the child is calm now").
- **State normal values out loud** for the child's age.
- **Always mention $SpO_2$**, even if the examiner did not provide an oximeter ("I would like to measure $SpO_2$").
- **Axillae**: examiners watch whether you listen there.

- **ENT last**, and **voice the epiglottitis rule** before touching a tongue depressor.
- **Finish with a classification** (WHO/IMCI) and two differentials.

# Mnemonics & memory aids

- **Order: "Look, Count, Measure, Touch, Listen, then Ears-Nose-Throat."**
- **WHO fast breathing "60–50–40"** (plus >30 at ≥5 years).
- **Severe distress "GHINC"**: **G**runting, **H**ead nodding, **I**ndrawing (severe), **N**asal flaring, **C**yanosis.
- **Pinna: "babies Down, big kids Up"** (both backwards).
- **Effusion vs collapse "push vs pull"**: effusion and pneumothorax **push** the trachea away; collapse **pulls** it towards.
- **Normal drum "Pearly, Light, Handle"**: pearly grey, light reflex, handle of malleus.
- **Node description "SCTMF"**: **S**ize/site, **C**onsistency, **T**enderness, **M**obility/matting, **F**luctuation (and skin).

# High-yield summary

## Numbers & doses to know

| Item | Value |
|---|---|
| Fast breathing (WHO) | **≥60** (<2 m), **≥50** (2–11 m), **≥40** (1–4 y), **>30** (≥5 y, UCG) |
| Severe tachypnoea | RR **≥70** |
| Count RR for | **60 seconds**, calm child |
| Hypoxaemia | $SpO_2$ **<90%** → oxygen; normal **≥95%** |
| Oximeter | Stable trace **30–60 s**; may over-read in dark skin |
| CRT | Normal **≤2 s**; abnormal **>3 s** |
| Normal RR | <1 y **30–40**; 1–2 y **25–35**; 2–5 y **25–30**; 5–12 y **20–25** |
| Normal HR | <1 y **110–160**; 1–2 y **100–150**; 2–5 y **95–140**; 5–12 y **80–120** |
| Minimum SBP (1–10 y) | **70 + 2 × age** |
| Nasal prong oxygen | **0.5–1** (neonate), **1–2** (infant), **2–4 L/min** (older) |
| Cervical node size | **≤1 cm** usually normal; >**2 cm** or any **supraclavicular** node needs investigation |
| Brodsky tonsils | 1+ <25%, 2+ 25–50%, 3+ 50–75%, 4+ >75% |
| Ear discharge | **<14 days** acute; **≥14 days** chronic (CSOM) |
| Peak flow | From about **5–6 years**; best of **3** |
| Apnoea (infant) | Pause **>20 s** (or shorter with bradycardia/cyanosis) |

## Classic exam traps

- **Guessing the RR** or counting for 15 s and multiplying: inaccurate; count 60 s.
- **Calling intercostal recession in a crying child "chest indrawing"**: must be lower chest wall, all the time, calm child.
- **Examining the throat in a drooling child**: dangerous.
- **Pulling the pinna up in an infant**: down and back for infants.
- **Calling transmitted upper-airway sounds "crackles"**: listen over the neck and after a cough.

- **Forgetting the axillae**: the right middle lobe and lingula are heard there.
- **Saying "asthma causes clubbing"**: it doesn't.
- **Diagnosing pneumonia from fast breathing in a wheezy or acidotic child** without a bronchodilator trial or malaria test.

### Questions seniors ask

**What is the respiratory rate and is it fast?** State the number counted over 60 s and the WHO cut-off for age ("54/min, fast for a 9-month-old: cut-off 50").

**Show me chest indrawing.** Point to the lower chest wall moving in on inspiration in a calm child, and contrast with intercostal recession.

**Why do you listen in the axillae?** The right middle lobe and lingula project there; consolidation can be missed from front and back alone.

**What would make you stop and call for help?** Stridor at rest with drooling, $SpO_2$ <90%, grunting/head nodding, silent chest, tracheal shift with shock, lethargy.

**What do you see in this eardrum?** Describe colour, translucency, position (bulging/retracted), light reflex, malleus, perforation/discharge, then name it (normal, AOM, OME, CSOM).

**How would you examine the throat of a 2-year-old?** On the parent's lap, arms and head held, torch, ask to open wide, tongue depressor briefly on the anterior tongue, look at tonsils, palate, uvula, pharynx, buccal mucosa; not at all if stridor/drooling.

**How do you tell an effusion from consolidation at the bedside?** Both are dull; effusion is stony dull with absent breath sounds and vocal resonance and may push the trachea away; consolidation has bronchial breathing, increased vocal resonance and crackles.

SELF-TEST

# Flashcards

27 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Recommended order of the paediatric respiratory and ENT examination. | **Observe** (end of bed, parent's lap) → **count RR (60 s) and $SpO_2$** → hands/face/neck → chest: **inspect, palpate, percuss, auscultate** → **ears → nose → throat last** → neck nodes → growth, heart, liver. |
| 2 | WHO fast-breathing cut-offs. | **≥60/min** (<2 months); **≥50** (2–11 months); **≥40** (12–59 months); **>30** (≥5 years, UCG 2023). Count a full minute, calm child. |
| 3 | How do you count RR accurately in an infant? | Child calm (not crying or feeding), chest/abdomen exposed, **watch the abdomen** (diaphragmatic breathing), count for a **full 60 seconds** with a timer; repeat if borderline. |
| 4 | Define lower chest wall indrawing. | The **lower chest wall goes IN when the child breathes IN**, present **all the time** in a **calm** child. Intercostal soft-tissue recession alone is not indrawing. |
| 5 | Signs of severe respiratory distress in a child. | **Grunting, head nodding, nasal flaring, severe chest indrawing, inability to drink/feed, central cyanosis**, RR ≥70; $SpO_2$ <90%. |
| 6 | Normal RR and HR by age (APLS). | RR: <1 y **30–40**; 1–2 y **25–35**; 2–5 y **25–30**; 5–12 y **20–25**. HR: <1 y **110–160**; 1–2 y **100–150**; 2–5 y **95–140**; 5–12 y **80–120**. |
| 7 | Pulse oximetry tips in children. | Correct probe size (toe/foot in infants), warm still limb, wait for a **steady pleth trace 30–60 s**, check pulse matches; note air vs $O_2$. May **over-read in dark skin** and poor perfusion. <90% = oxygen. |
| 8 | Stridor vs wheeze. | **Stridor**: harsh, **inspiratory**, extrathoracic (larynx/trachea), loudest over the neck — croup, epiglottitis, FB. **Wheeze**: musical, **expiratory**, intrathoracic airways — asthma, bronchiolitis. |
| 9 | When must you NOT examine the throat? | **Stridor, drooling, tripod/sniffing posture, muffled voice, toxic look** (suspected epiglottitis/retropharyngeal abscess/severe croup): risk of complete obstruction. Keep upright with parent, call airway help. |
| 10 | How to straighten the ear canal for otoscopy. | **Infants: pull the pinna down and back. Older children: up and back.** Largest speculum that fits; hold otoscope like a pen; **brace the little finger on the child's head**. |
| 11 | Appearance of a normal tympanic membrane. | **Pearly grey, translucent**, slightly concave, **cone of light anteroinferiorly**, **handle of malleus** visible, mobile on pneumatic otoscopy. |
| 12 | Otoscopic signs of AOM vs OME. | **AOM**: red/yellow, opaque, **bulging**, landmarks and light reflex lost (± pus if perforated). **OME (glue ear)**: dull, **retracted**, prominent malleus, **fluid level/bubbles**, reduced mobility, no acute illness. |
| 13 | CSOM: definition and otoscopic findings. | Ear discharge **≥14 days** through a **perforation** (central or attic). Attic perforation with white debris and foul smell = **cholesteatoma** → ENT. |
| 14 | Brodsky tonsil grades. | 0 in fossa/removed; **1+ <25%**; **2+ 25–50%**; **3+ 50–75%**; **4+ >75%** of the oropharyngeal width ('kissing tonsils'). |
| 15 | How to examine a toddler's throat safely. | Parent holds child on lap facing forward (one arm across the child's arms/chest, hand on forehead); torch; ask to 'open wide/roar'; tongue depressor briefly on the **anterior two-thirds** of the tongue. Look at tonsils, uvula, palate, pharynx, buccal mucosa. Never with stridor/drooling. |

| # | Question | Answer |
|---|---|---|
| 16 | Throat findings and their meaning. | Exudate + tender anterior nodes = **GAS**; grey adherent membrane that bleeds = **diphtheria**; uvula pushed away + trismus = **quinsy**; white spots opposite molars = **Koplik (measles)**; white plaques = **thrush (HIV/antibiotics)**. |
| 17 | Order of palpating head and neck lymph nodes. | **Submental → submandibular → tonsillar (jugulodigastric) → anterior cervical → posterior cervical → supraclavicular → pre-auricular → post-auricular → occipital.** |
| 18 | Which cervical nodes need investigation in a child? | Nodes **>2 cm**, any **supraclavicular** node, **hard/fixed/rapidly growing**, **matted/fluctuant or with a sinus** (TB), persisting >4–6 weeks, or with weight loss/night sweats/hepatosplenomegaly. ≤1 cm soft mobile nodes are common and normal. |
| 19 | Causes of clubbing in a Ugandan child. | **Bronchiectasis** (post-measles, pertussis, TB), **HIV lung disease (LIP)**, **cyanotic congenital heart disease**, empyema/lung abscess, **cystic fibrosis**, IBD, liver cirrhosis, endocarditis. **Not asthma.** |
| 20 | Pleural effusion vs consolidation vs collapse vs pneumothorax at the bedside. | **Effusion**: stony dull, absent sounds, trachea **pushed away**. **Consolidation**: dull, **bronchial breathing**, crackles, ↑vocal resonance. **Collapse**: dull, ↓sounds, trachea **pulled towards**. **Pneumothorax**: **hyper-resonant**, absent sounds, trachea pushed away. |
| 21 | Transmitted upper-airway sounds: how to recognise them. | **Loudest over the neck/trachea**, **symmetrical**, coarse, **change or clear after a cough or nasal suction**. Don't call them crackles or wheeze. |
| 22 | Why auscultate in the axillae? | The **right middle lobe** and **lingula** project there; consolidation in these lobes can be missed if you listen only front and back. |
| 23 | Deep, fast breathing with a clear chest — what should you think of? | **Metabolic acidosis**: severe malaria, DKA, severe dehydration, sepsis, salicylate poisoning — not pneumonia. Check malaria test, glucose, hydration. |
| 24 | Rinne and Weber in a conductive hearing loss (e.g. right glue ear). | **Rinne right negative** (bone > air); **Weber lateralises to the right** (affected) ear. Use a 512 Hz fork, older children only. |
| 25 | Anterior rhinoscopy: what are you looking for? | Push the tip up, look with otoscope/torch: **septum** (deviation, haematoma), **turbinates** (pale/swollen = allergy; red = infection), **polyps** (think **CF** in a child), **foreign body** (unilateral foul discharge), discharge. |
| 26 | Nasal prong oxygen flows by age. | **Neonates 0.5–1 L/min; infants 1–2 L/min; older children 2–4 L/min.** Target $SpO_2$ ≥90% (≥94% with emergency signs). |
| 27 | Minimum acceptable systolic BP in a child 1–10 years. | **70 + (2 × age in years)** mmHg. Use a cuff whose bladder covers 80–100% of the arm circumference. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** You are asked to count the respiratory rate of a 7-month-old girl who is crying loudly on her mother's lap. What is the BEST way to obtain an accurate rate?

A Wait until she is calm (e.g. after breastfeeding stops or while held), expose the abdomen and count for a full 60 seconds
B Count for 15 seconds while she cries and multiply by 4
C Use the heart rate on the pulse oximeter divided by 3
D Count only while she is breastfeeding because she is quiet
E Estimate by looking for 10 seconds; experienced clinicians do not need to count

**2.** A 14-month-old boy (9 kg) has cough and fever. While calm on his mother's lap his RR is 46/min; the lower chest wall goes in with each breath. $SpO_2$ 95%, alert, drinking. Malaria RDT negative. How should you classify him?

A No pneumonia: cough or cold, because RR is below 50
B Severe pneumonia, because chest indrawing is present
C Pneumonia (fast breathing and chest indrawing, no danger signs)
D Bronchiolitis, because he is under 2 years
E Asthma, because indrawing indicates bronchospasm

**3.** A 4-year-old girl is brought in sitting upright, leaning forward with her mouth open, drooling, with a soft inspiratory stridor and a muffled voice. Your OSCE examiner hands you a tongue depressor and torch. What should you do?

A Examine the throat quickly to confirm the diagnosis before calling for help
B Lie her flat to examine the throat more easily
C Examine the throat but avoid the posterior tongue
D Ask the mother to hold her down firmly so you can look
E Decline to examine the throat, keep her calm and upright with her mother, give oxygen without distress, and call for senior anaesthetic/ENT help

**4.** You are about to examine the ear of a 10-month-old infant with an otoscope. Which technique is CORRECT?

A Use the smallest speculum and pull the pinna up and back
B Use the largest speculum that fits, pull the pinna down and back, and brace your hand on the child's head
C Hold the otoscope like a hammer with no contact with the head so that it does not move
D Lie the infant flat and insert the speculum until resistance is felt
E Pull the pinna forward to open the canal

**5.** A 15-month-old with 2 days of fever and irritability has a right tympanic membrane that is red, opaque and bulging, with no light reflex and the malleus not visible. The left drum is pearly grey with a normal light reflex. What is the diagnosis?

A Otitis media with effusion (glue ear)
B Otitis externa
C Chronic suppurative otitis media
D Right acute otitis media
E Normal findings in a crying child

**6.** A 7-year-old boy (20 kg) has 3 weeks of fever and a TB contact. The trachea is deviated to the left. The right lower chest is stony dull with absent breath sounds and absent vocal resonance; bronchial breathing is heard just above the dull area. What is the MOST likely finding?

- A Right lower lobe collapse
- B Right pneumothorax
- C Right pleural effusion
- D Right lower lobe consolidation only
- E Normal chest with transmitted sounds

**7.** A 9-month-old with a runny nose has coarse noises all over the chest on auscultation. RR 36/min, no indrawing, $SpO_2$ 98%, feeding well. The noises are loudest over the neck, sound the same on both sides, and disappear after the nose is suctioned. What are they?

- A Fine crackles of pneumonia
- B Polyphonic wheeze of asthma
- C Pleural rub
- D Stridor of croup
- E Transmitted upper-airway sounds

**8.** A 6-year-old girl (18 kg) is breathing deeply and fast (RR 44/min) with a clear chest, no indrawing and $SpO_2$ 98%. She has had fever for 3 days and is drowsy. Her conjunctivae are pale. What is the MOST appropriate next step?

- A Malaria test (RDT/smear), blood glucose and haemoglobin; consider severe malaria with acidosis
- B Oral amoxicillin for pneumonia
- C Nebulised salbutamol
- D Chest X-ray before any treatment
- E Throat swab for streptococcus

**9.** A 3-year-old boy has a firm, matted, non-tender 3 cm swelling in the right posterior cervical region with a discharging sinus. He has lost weight over 2 months and his uncle has TB. What is the MOST appropriate next step?

- A Oral amoxicillin-clavulanate for 10 days and review
- B Reassure; reactive nodes are common in children
- C Incision and drainage in OPD and send home
- D Fine-needle aspiration of the node for Xpert/cytology, with CXR and HIV test
- E Start chemotherapy for lymphoma

**10.** During bedside teaching, a 5-year-old girl with asthma (19 kg) who was wheezing loudly an hour ago is now drowsy, the chest is almost silent on auscultation, RR 18/min, $SpO_2$ 86% on a face mask. What does this mean?

- A The asthma has resolved and she is sleeping
- B Life-threatening asthma with exhaustion: call for help now and escalate treatment
- C She has developed pneumonia and needs antibiotics only
- D The oximeter is faulty because the chest is clear
- E Salbutamol toxicity; stop bronchodilators

**11.** A 2-year-old has a wet cough. On inspection the nails show loss of the angle at the nail fold with increased curvature. Which condition is LEAST likely to explain this sign?

A Bronchiectasis after measles
B Lymphoid interstitial pneumonitis (HIV)
C Cyanotic congenital heart disease
D Cystic fibrosis
E Asthma

**12.** A 4-year-old girl has had painless discharge from the left ear for 6 weeks. After dry mopping, otoscopy shows a central perforation of the left drum with pus in the middle ear. What is the BEST management to start in OPD?

A Oral amoxicillin for 5 days
B Cotton wool plug soaked in warm oil
C Teach dry wicking 3 times daily and ciprofloxacin ear drops for 2 weeks; HIV test and hearing assessment
D Urgent mastoidectomy
E Syringe the ear with water daily and allow swimming

**13.** A 2-year-old boy has a 3-week history of a foul-smelling, blood-stained discharge from the right nostril only. He is otherwise well with no fever. What is the MOST likely diagnosis?

A Acute bacterial sinusitis
B Nasal foreign body
C Allergic rhinitis
D Common cold
E Nasal polyps from cystic fibrosis

**14.** A 6-year-old's oximeter reading fluctuates between 85% and 97% while she wriggles, and the plethysmograph trace is irregular. Her lips are pink and she is breathing comfortably at 22/min. What should you do?

A Calm her, warm the hand, reposition or change the probe, keep the limb still and wait for a steady waveform before recording
B Record the lowest value and start oxygen
C Record the highest value and move on
D Average the two numbers and record 91%
E Ignore oximetry in children; it is unreliable

**15.** In an OSCE you are asked to examine the throat of a 3-year-old with fever and sore throat who has no stridor or drooling. Which technique is MOST appropriate?

A Lie the child flat on the couch and press the tongue depressor firmly on the back of the tongue
B Ask the child to stand alone and open the mouth while you hold the head
C Insert the tongue depressor as far as possible to see the posterior pharynx
D Seat the child on the parent's lap with arms and head gently held, use a torch, and place the depressor briefly on the anterior two-thirds of the tongue
E Skip the throat; it is never necessary in children

## Answers & explanations

**1. A** · Wait until she is calm (e.g. after breastfeeding stops or while held), expose the abdomen and count for a full 60 seconds

A is right: WHO/IMCI requires counting for a **full minute** in a **calm** child, watching the abdomen in infants. B is wrong: crying raises RR, and 15-second counts multiply errors. C is wrong: HR:RR ratios are not a valid way to measure RR. D is wrong: feeding changes the breathing pattern; count when she is calm but not feeding. E is wrong: estimates are frequently wrong and are the commonest cause of misclassification.

**2. C** · Pneumonia (fast breathing and chest indrawing, no danger signs)

C is right: at 12–59 months the cut-off is **≥40**, so RR 46 is fast; with **lower chest wall indrawing** and no danger sign or hypoxaemia this is WHO **pneumonia** (oral amoxicillin). A is wrong: the 50 cut-off applies to 2–11 months. B is wrong: since WHO 2014, indrawing without danger signs is 'pneumonia', not severe pneumonia. D is wrong: bronchiolitis needs a coryzal prodrome with wheeze/crackles; age alone doesn't make the diagnosis. E is wrong: indrawing reflects stiff lungs or obstruction of any cause; there is no wheeze here.

**3. E** · Decline to examine the throat, keep her calm and upright with her mother, give oxygen without distress, and call for senior anaesthetic/ENT help

E is right: drooling, tripod posture, muffled voice and stridor suggest **epiglottitis**; throat examination, lying flat or restraint can cause **complete airway obstruction**. Say this aloud and call the airway team. A, C and D are wrong: any throat examination or restraint risks obstruction. B is wrong: lying flat lets the swollen epiglottis fall back.

**4. B** · Use the largest speculum that fits, pull the pinna down and back, and brace your hand on the child's head

B is right: in infants the canal is straightened by pulling the pinna **down and back**; the **largest speculum that fits** gives the best view, and **bracing** the hand means the otoscope moves with the head, preventing injury. A is wrong: up and back is for older children, and too small a speculum gives a poor view and goes too deep. C is wrong: an unbraced otoscope injures the canal if the child moves. D is wrong: inserting until resistance risks canal and drum trauma. E is wrong: pulling forward does not straighten the canal.

**5. D** · Right acute otitis media

D is right: a **red, bulging drum with loss of landmarks** in a febrile, irritable child = **acute otitis media**. A is wrong: glue ear shows a dull, **retracted** drum with fluid/bubbles and no acute illness. B is wrong: otitis externa affects the canal (swelling, pain on moving the pinna) with a normal drum. C is wrong: CSOM has a perforation with discharge ≥14 days. E is wrong: crying can redden a drum but does not make it **bulge** with lost landmarks.

**6. C** · Right pleural effusion

C is right: **stony dullness, absent breath sounds and vocal resonance**, with the trachea **pushed away** and bronchial breathing above the fluid (compressed lung), = **pleural effusion** (TB effusion vs empyema → CXR/ultrasound and tap). A is wrong: collapse **pulls** the trachea towards the affected side. B is wrong: pneumothorax is **hyper-resonant**. D is wrong: consolidation has bronchial breathing and **increased** vocal resonance over the dull area, without tracheal shift. E is wrong: these are clear abnormal signs.

**7. E** · Transmitted upper-airway sounds

E is right: sounds that are **loudest over the neck, symmetrical** and **clear after nasal suction or coughing** are **transmitted upper-airway sounds** from nasal secretions; with a normal RR and $SpO_2$ this is a cold. A is wrong: crackles are localised to the lung, inspiratory and don't clear with suction. B is wrong: wheeze is musical and expiratory, from intrathoracic airways. C is wrong: a pleural rub is a localised creak in both phases. D is wrong: stridor is a harsh inspiratory noise from the larynx with a barking cough.

**8. A** · Malaria test (RDT/smear), blood glucose and haemoglobin; consider severe malaria with acidosis

A is right: **deep, fast (acidotic) breathing with a clear chest** in a febrile, drowsy, pale child in Uganda suggests **severe malaria with metabolic acidosis** (or severe anaemia): test for malaria, glucose and Hb and treat (IV artesunate, transfusion if needed). B is wrong: there are no chest signs of pneumonia; this pattern is respiratory compensation for acidosis. C is wrong: there is no wheeze. D is wrong: CXR delays urgent treatment and is unlikely to explain the picture. E is wrong: the throat is not the issue.

**9. D** · Fine-needle aspiration of the node for Xpert/cytology, with CXR and HIV test

D is right: a **firm, matted node with a sinus**, weight loss and a TB contact = **TB lymphadenitis** until proven otherwise: **FNA for Xpert Ultra/cytology**, CXR and HIV test, then treat (peripheral node TB is non-severe → 4-month regimen if confirmed). A is wrong: antibiotics delay diagnosis. B is wrong: a 3 cm matted node with a sinus is not a simple reactive node. C is wrong: I&D of TB nodes causes chronic sinuses and doesn't diagnose. E is wrong: lymphoma needs tissue diagnosis first, and the picture favours TB.

**10. B** · Life-threatening asthma with exhaustion: call for help now and escalate treatment

B is right: a **silent chest**, **drowsiness**, a **falling RR** and **low $SpO_2$** = **life-threatening asthma with exhaustion**: stop teaching, call for help, back-to-back salbutamol + ipratropium, IV magnesium, prepare for ventilation. A is wrong: a quiet chest here means no airflow. C is wrong: nothing suggests pneumonia, and antibiotics wouldn't address the emergency. D is wrong: the clinical picture fits the low $SpO_2$. E is wrong: toxicity causes tachycardia and tremor, not a silent chest.

**11. E** · Asthma

E is right: **asthma does not cause clubbing**; a clubbed asthmatic child needs evaluation for another diagnosis. A is wrong: bronchiectasis (post-measles, pertussis or TB) is a classic cause. B is wrong: LIP in HIV-positive children causes clubbing. C is wrong: cyanotic heart disease causes clubbing. D is wrong: cystic fibrosis causes clubbing through chronic suppurative lung disease.

**12. C** · Teach dry wicking 3 times daily and ciprofloxacin ear drops for 2 weeks; HIV test and hearing assessment

C is right: discharge **≥14 days** with a **perforation** = **CSOM**: dry wicking plus topical quinolone drops for 2 weeks (UCG/IMCI), keep water out, HIV test and hearing check. A is wrong: systemic antibiotics are not useful in CSOM. B is wrong: plugs and oil keep the ear wet and infected. D is wrong: there are no signs of mastoiditis. E is wrong: water entry worsens CSOM.

**13. B** · Nasal foreign body

B is right: a **unilateral, foul-smelling (often blood-stained) nasal discharge** in a toddler is a **foreign body** (bead, bean, foam, groundnut) until proven otherwise; look with a light and remove or refer to ENT (urgently if a button battery). A is wrong: sinusitis is usually bilateral and maxillary sinuses are small at 2 years. C is wrong: allergic rhinitis gives clear bilateral discharge with sneezing. D is wrong: a cold is bilateral and resolves in about a week. E is wrong: polyps are bilateral and rare at this age.

**14. A** · Calm her, warm the hand, reposition or change the probe, keep the limb still and wait for a steady waveform before recording

A is right: a reliable reading needs a **steady pleth waveform** for 30–60 s, a still, warm limb and the right probe; motion causes false readings. B is wrong: a motion artefact is not hypoxaemia, and she looks well, though you should confirm. C and D are wrong: recording an arbitrary or averaged value from an unreliable trace misleads the team. E is wrong: pulse oximetry is essential and reliable when done properly.

**15. D** · Seat the child on the parent's lap with arms and head gently held, use a torch, and place the depressor briefly on the anterior two-thirds of the tongue

D is right: the parent's lap with gentle restraint, a torch, and a **brief** depression of the **anterior two-thirds** of the tongue gives a good view with minimal gagging. A and C are wrong: pressing on the posterior tongue causes gagging and vomiting, and lying flat is unnecessary. B is wrong: a standing 3-year-old will pull away; positioning with the parent is safer. E is wrong: the throat is a common source of fever (tonsillitis, diphtheria, measles, thrush) and should be examined when safe.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Bedside teaching: a toddler with fever and cough

During Dr Mawanda's bedside teaching on the Mengo paediatric ward, you are asked to examine Amina, a 2-year-old girl (11 kg) admitted this morning with 3 days of fever and cough. She is sitting on her mother's lap, quiet but alert.

**1. How do you start, and what do you do before touching her?**

> Hand hygiene, introduce yourself, explain and get consent; keep her on mum's lap, expose the chest, keep warm. **Observe** first: alertness, colour, noises, posture, flaring, indrawing, head nodding, drooling, feeding. Then **count RR for 60 s** while calm and **measure $SpO_2$** with the right probe.

**2. RR 48/min, $SpO_2$ 93% in air, HR 136, T 38.6 °C, nasal flaring, lower chest wall indrawing, no grunting or head nodding. Interpret.**

> **Fast breathing** (cut-off ≥40 at 12–59 months) and **chest indrawing**; no danger sign; $SpO_2$ 93% (not hypoxaemic by the <90% threshold but monitor closely). WHO class so far: **pneumonia**.

**3. Chest: trachea central, reduced expansion on the right, dull percussion right base posteriorly, bronchial breathing and fine crackles there, left clear, no wheeze. What do you conclude?**

> **Right lower lobe consolidation** (dullness, bronchial breathing, crackles, no tracheal shift). Listened in the **axillae** too (to exclude middle lobe). No wheeze, so no bronchodilator trial needed.

**4. Describe how you would examine her ears and throat, and what you would look for.**

> Ears: largest speculum that fits, pinna **down and back** (infants and young toddlers; up and back in older children), brace hand; look for red bulging drum (AOM). Throat (no stridor/drooling): mum holds her, torch, brief depressor on anterior tongue; look for exudate, membrane, Koplik spots, thrush, uvula position. Then palpate nodes in order.

**5. Present your findings and plan in 30 seconds.**

> "Amina, 2 years, 11 kg, WAZ −0.8, fully immunised, HIV-negative, 3 days of fever and cough; RR 48 (fast), indrawing, $SpO_2$ 93% in air, signs of right lower lobe consolidation; ENT normal; no danger signs. **WHO pneumonia, right lower lobe.** Plan: amoxicillin 40 mg/kg (440 mg) twice daily × 5 days, paracetamol, monitor $SpO_2$ and RR, review in 2–3 days; admit if danger signs or $SpO_2$ <90%."

**Take-away:** Observe, count and measure before you listen; then localise, check the ENT, and finish with a WHO classification and plan.

## Case 2: Noisy breathing at night

A 15-month-old boy (10 kg) is brought to emergency at 2 a.m. with noisy breathing. His mother imitates a 'barking' cough. He is on her lap, crying intermittently.

**1. What is the first thing you need to establish about the noise, and how?**

> Whether it is **stridor** (inspiratory, harsh, from the upper airway: listen near the mouth/neck) or **wheeze** (expiratory, musical, from the chest), and whether stridor is present **at rest when calm**. Also look for drooling, tripod posture, toxicity and cyanosis.

**2. He has inspiratory stridor when calm, moderate suprasternal and subcostal recession, $SpO_2$ 95%, no drooling, drinking, T 37.9 °C. Which parts of the examination will you avoid?**

> Avoid the **throat examination with a spatula**, lying him flat, and painful procedures; avoid separating him from his mother. Examine chest and ears gently on her lap.

**3. Classify and give the first treatment.**

> **Moderate croup** (stridor at rest, recession, alert, no drooling). **Dexamethasone 0.15–0.6 mg/kg orally** once (1.5–6 mg), keep calm, observe $SpO_2$ and work of breathing.

**4. Thirty minutes later the stridor is quieter but he is drowsy, more cyanosed and $SpO_2$ is 86%. What does this mean?**

> **Impending respiratory failure**: a softening stridor with drowsiness and cyanosis means less air is moving, not improvement. Oxygen, **nebulised adrenaline 1:1000, 5 mL** (0.5 mL/kg, max 5 mL), call the anaesthetist for airway support.

**5. How would the examination have differed if this were epiglottitis?**

> Toxic, high fever, **drooling**, **tripod/sniffing** posture, muffled voice, little cough, refusing to swallow. You would **not** examine the throat or lie him down; you'd keep him upright with his mother and get the airway team to theatre.

**Take-away:** Identify stridor vs wheeze, judge it at rest, protect the airway by not examining the throat, and remember that quieter stridor with drowsiness is deterioration.

## Case 3: Fever without an obvious source

A 10-month-old girl (8 kg) has had fever for 2 days and is irritable and pulling at her right ear. Malaria RDT is negative. Her mother says she has had 'a cold' this week.

**1. Which danger signs and vital signs do you check first?**

> Able to breastfeed? Vomiting everything? Convulsions? Lethargy? Then **RR (60 s)** (cut-off ≥50 at 2–11 months), **$SpO_2$**, HR, temperature, CRT; fontanelle and neck stiffness (meningitis).

**2. She is breastfeeding, alert, RR 40, $SpO_2$ 98%, fontanelle flat, chest clear. How do you examine her ears?**

> Mother holds her sideways on the lap, one arm around her body and arms, the other hand holding her head to the chest. Largest speculum that fits, **pinna down and back** (infant), little finger braced on her head; inspect the canal and drum on both sides (good ear first).

**3. Right drum: red, opaque, bulging, no light reflex. Left drum normal. No discharge, no mastoid swelling. Diagnosis and treatment?**

> **Right acute otitis media.** Amoxicillin 40 mg/kg = **320 mg twice daily for 5 days** (WHO/IMCI; UCG 15 mg/kg 8-hourly), paracetamol 120 mg (15 mg/kg) 6-hourly, review in 5 days.

**4. What would make you admit her urgently?**

> Swelling, redness or tenderness **behind the ear** with the pinna pushed forward (**mastoiditis**), neck stiffness/bulging fontanelle or convulsions (**meningitis**), facial palsy, danger signs, or hypoxaemia.

**5. Five weeks later she has a painless discharge from the same ear for 3 weeks. What do you find and do?**

> Likely **CSOM** (discharge ≥14 days): after dry mopping, a perforation. **Teach wicking 3 times daily**, ciprofloxacin ear drops for 2 weeks, keep water out, **HIV test**, hearing assessment, ENT referral if not dry.

**Take-away:** In a febrile infant always look in the ears after checking danger signs, RR and $SpO_2$; technique (pinna down and back, braced otoscope) gives you the diagnosis safely.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.